Automated Detection of Answerhood in Presidential Q&A: A Hierarchical Annotation and Classification Approach

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Abstract

This paper presents a pilot study of automated answerhood detection in presidential debate transcripts. A three-layer annotation scheme has been developed to distinguish between (1) whether a response is direct or not direct, (2) the type of evasion strategy employed, and (3) the rhetorical mechanism used by a speaker. Six individuals annotated a dataset of 212 question-answer pairs, yielding a corpus of 390 instances, the annotators, consisted of three graduate students and three high school seniors working from a shared set of annotation guidelines. Inter-annotator agreement (IAA) was calculated using Krippendorff's alpha and pairwise agreement was calculated using Cohen's kappa, revealing moderate reliability on Layer 1 (α = 0.4471) but poor reliability for on Layer 2 (α = 0.1479), this indicates that evasion categories require label refinement. Logistic regression classifiers via TF-IDF trained with balanced class weights achieve a macro F1 average of 0.70 and a weighted average of 0.72 for Layer 1, 0.49 and 0.50 for Layer 2, and 0.35 and 0.47 for Layer 3. Classification performance on all layers closely correlates with annotator agreement, suggesting that label noise constitutes a primary bottleneck for this study.

1 Introduction

Answerhood, the degree to which a response addresses the question posed, is a central concern in pragmatics and political discourse analysis. Politicians routinely deflect, re-frame, or outright ignore questions in ways that are immediately perceived by human listeners, yet is rather difficult for computational models to grasp. The question-answer pair (Schefloff and Sacks, 1973) provides a theoretical baseline for analysis, whereby unless the response given resolves the question posed, a thorough review is warranted.

Presidential debates are a rich source for studying evasion and non-answers. The constraints of a formal debate perfectly align with general experimental design requirements. The premise of a debate is such that they pressure participants to provide resolution to posed questions, providing the perfect conditions for collection of direct resolution and or avoidance strategies on part of the participant.

This study makes four contributions. First, it developed a presidential debate dataset. Second, it proposes a hierarchical three-layer annotation scheme for answerhood in political Q&A, grounded in previous work on question evading (Rasiah, 2010) and discourse coherence (Roberts, 2019). Third, it reports IAA across six annotators of mixed expertise and experience levels, revealing the relative ease of fine-grained judgments and the considerable difficulties involving fine-grained evasion classification. Fourth, it establishes a computational baseline using TF-IDF features and balanced logistic regression to evaluate performance per layer and discusses that relationship between annotation reliability and classifier performance.

2 Related Work

The study of answerhood exists where pragmatics, conversation analysis, and general political communication converge. Within the field of pragmatics, relevance theory (Sperber, 1986) and Gricean maxims (Grice, 1975) are a commonly cited framework for dialogue whereby a non-answer will violate the maxim of relation by failing to provide information the questioner has requested. Some discourse analysts have focused on the structure of discourse in the form of question-answer pairs and have noted that ill-preferred second pair parts are commonly associated with delays, accounts, and topic shifts (Schegloff, 2007).

Studies on political discourse have identified several strategies employed by politicians to avoid answering questions directly. Rasiah (2010) iden-

tifies several strategies; topic shifts, blame attribution, and narrative deflection as consistent strategies used by members of parliament when answering questions. Clayman (2001) observed how interviewees resist answering questions by attacking the questioner's framing rather than engaging in debate. These strategies for the foundation for layers 2 and 3 labels of the annotation scheme.

Question answering research has largely focused on fact-based retrieval Voorhees (2001) and reading comprehension rather than whether a participant has addressed a posed question.

3 Data and Annotation

3.1 Corpus

The corpus consists of question-answer pairs drawn from the 2000 and 2004 U.S. presidential debate transcripts acquired from the The American Presidency Project. The dataset is comprised of 390 annotated instances (the entire corpus yields 212 Q&A pairs). Each instance consists of a moderator/journalist/audience member question paired with the following candidate. Input texts for annotation and classification were formed by concatenating the question and the candidate utterance with a [SEP] delimiter, following standard practices in dual-sequence classification as per (Devlin et al., 2019)

3.2 Annotation Scheme

This paper adopts a hierarchical annotation scheme comprised of three layers, each layer depending on the label assigned to the preceeding layer. This design reflects the theoretical claim that evasion strategies (layer 2) and rhetorical mechanism (layer 3) are only meaningful when a response has been identified as not direct (layer 1).

3.2.1 Layer 1: Directness (*DIRECT* or *NOT_DIRECT*)

Annotators labeled each response as *DIRECT* (the candidate addresses the posed question) or *NOT_DIRECT* (the candidate does not address the posed question). This is a binary single label classification task.

3.2.2 Layer 2: Evasion Strategy (*NOT_DIRECT only*)

Annotators select one or more of the following labels as they see fit according to candidate's response. *TOPIC_SHIFT* (redirecting to a preferred or alternative topic), *PASS_BLAME* (deflecting responsibility), or *STORY_TELLING* (using narratives to avoid directly addressing the posed question).

3.2.3 Layer 3: Rhetorical Mechanism (*NOT_DIRECT* only)

Annotators select from *ATTACK_QUESTIONER*, *ATTACK_OPPONENT*, or *DEFERRED_RESPONSE*.

3.3 Annotators

Six individuals served as annotators: three graduate students and three high school seniors. Annotators were provided with a set of annotation guidelines prior to labeling. Due to time constraints, no formal annotator training session, calibration round, or inter-annotator disagreement resolution procedure was conducted. Annotation followed a partial overlap design in which each annotator labeled a subset o the corpus. No single item was rated by all six annotators

3.4 Data Quality

One data quality issue was identified and resolved prior to analysis. Six instances in layer 3 bore the bare label *ATTACK* rather than the fully specified *ATTACK_QUESTIONER* or *ATTACK_OPPONENT*, this was a result of a faulty spreadsheet template with three of the annotators. These instances could then not therefore be reliably assigned to either category and were excluded from layer 3 analysis in order to not contaminate said analysis.[1]

4 Inter-Annotator Agreement

Because the annotation followed a partial overlap design, Krippendorff's alpha (α) is reported as the primary agreement metric, as it was specifically designed to handle missing observations (Krippendorff, 2004). For annotator pairs sharing at least two items, pairwise Cohen's kappa (κ) is reported. Table 1 summarizes agreement across all three layers.[2]

[1] One row containing no valid layer 1 label was also removed prior to training.

[2] Krippendorff alpha thresholds follow Landis and Koch (1977): α < 0.20 = poor; 0.20 - 0.40 = fair; 0.41 - 0.60 = moderate; > 0.60 = substantial

Table 1: Inter-annotator agreement by layer

Layer	Krippendorff α	Pairwise κ (range)	Overlapping Items	Interpretation
Layer 1 (Directness)	0.4471	0.25 – 0.66	79 items ($\geq$ 2 raters)	Moderate
Layer 2 (Evasion)	0.1479	-0.12 – 0.58	32 items ($\geq$ 2 raters)	Poor
Layer 3 (Rhetorical Device)	0.3558	0.00 – 1.00	31 items ($\geq$ 2 raters)	Fair

5 Method

5.1 Layer 1

Layer 1 achieved moderate agreement ($\alpha = 0.4471$), suggesting that the coarse binary distinction between direct and non-direct responses is reliably perceptible even without formal annotator training. Pairwise kappa ranges from $\kappa = 0.25$ (AM vs. BD)[3] to $\kappa = 0.66$ (MR vs. TM; CLM vs. TM). The wide range in values likely reflects differences in background knowledge and familiarity with political discourse across the annotator pool. This is likely do to the mixed composition of the annotators (high school vs. graduate level).

5.2 Layer 2

Layer 2 agreement is poor ($\alpha = 0.1479$), with two pairwise kappas at or below zero [4]. This indicates that annotators wee unable to consistently distinguish among *TOPIC_SHIFT*, *PASS_BLAME*, *STORY_TELLING*. This is consistent with the absence of a calibration session prior to annotation. Without shared worked examples and discussion of boundary cases, fine-grained evasion categories are difficult to gauge reliably.

5.3 Layer 3

Layer 3 shows fair overall agreement among annotators ($\alpha = 0.3558$), though pairwise kappas once again vary substantially. The near perfect agreement between one annotator pair ($\kappa = 1.00$, n=12) should be interpreted cautiously given the small number of shared samples.

6 Approach

6.1 Features and Model

Each instance is represented as a concatenated question-answer string separated by [SEP]. Text is vectorized using TF-IDF with unigram and bigram features with a minimum document frequency of 2, and a maximum document frequency of 0.9. Layer 1 uses standard multinomial logistic regression. Layers 2 and 3 use One-vs-Rest logistic regression to accommodate multi-label outputs. All classifiers use the stochastic optimization via sklearn's SAGA solver with *class_weight='balanced'* to mitigate class imbalance, and a maximum of 3,000 iterations to ensure convergence.

6.2 Experimental Setup

The train/test split was 80/20, stratified on layer 1 labels. The same indices are used for all three layers. Layers 2 and 3 are trained and evaluated exclusively on *NOT_DIRECT* instances (121 training, 30 test). For the test set, layer 1 achieves a majority-class baseline accuracy of 60.526%

7 Results

Table 2 reports precision, recall, and F1 scores per class across all three layers. The balanced classifiers represent a substantial improvement over unweighted models, which predicted only the majority class for layers 2 and 3.[5]

Layer 1 achieves a macro F1 = 0.70, substantially above the majority baseline. Precision between DIRECT (0.73) and NOT_DIRECT (0.75) is relatively balanced, recall for NOT_DIRECT remains lower (0.50), reflecting the slight class imbalance (0.02).

Layer 2 achieves a macro F1 = 0.49. The balanced classifier now identifies all three evasion strategies, this is a substantial improvement over the unweighted model which collapsed to a majority-class prediction as a result of the unbalanced classes. Performance is moderate across

[3] Annotator AM vs. BD

[4] $\kappa = 0.0000$ (CLM vs.MR), $\kappa = -0.0687$ (CLM vs. TM)

[5] The majority-class baseline (always predicting DIRECT) achieves 60.526% accuracy on the layer 1 test set. The balanced classifier achieves 74% accuracy

Table 2: Classification results by layer and class

Layer	Class	Precision	Recall	F1	Support
Layer 1 (Directness)	DIRECT	0.73	0.89	0.80	46
	NOT_DIRECT	0.75	0.50	0.60	30
	Macro avg	0.74	0.70	0.70	76
Layer 2 (Evasion)	TOPIC_SHIFT	0.50	0.62	0.55	13
	PASS_BLAME	0.38	0.38	0.38	8
	STORY_TELLING	0.067	0.44	0.53	9
	Macro avg	0.51	0.48	0.49	30
Layer 3 (Rhetorical Device)	ATTACK_QUESTIONER	0.00	0.00	0.00	1
	ATTACK_OPPONENT	0.55	0.67	0.60	9
	DEFERRED_RESPONSE	0.50	0.33	0.40	12
	Macro avg	0.35	0.33	0.33	22

categories, consistent with the fair to poor IAA on this layer.

Layer 3 achieves a macro F1 = 0.33. ATTACK_QUESTIONER yields an F1 = 0.00, this is expected given that only one test instance bears this label. ATTACK_OPPONENT achieves an F1 = 0.60 and DEFERRED_RESPONSE yields an F1 = 0.40, these showcase meaningful above average change results.

8 Discussion

The results of the study reveal a consistent relationship between IAA and classifier performance across all three layers. Layer 1 is the only layer with moderate agreement, likely due to its binary nature. Layers 2 and 3 show poor and fair agreement respectively, and yield correspondingly low F1 scores. This pattern is consistent with the theoretical claim that label noise constitutes an upper bound on supervised classification performance.

The annotator pool, being comprised of such varying annotators, coupled with the absence of a training session is a plausible contributor to the varying results in the pairwise kappa observed in layer 1 (0.25 - 0.66). Annotators more familiar with political rhetoric and pragmatic theory may interpret the DIRECT/NOT_DIRECT distinction differently than others without such prior knowledge. The absence of a calibration session means that these differences were not resolved before annotation began.

Substantial improvements wee made in layers 2 and 3 following the implementation of balanced class weights. This improvement illustrates the importance of balancing classes for any dataset, but of significant importance in small corpora. Unweighted classifiers in the presence of rare minority classes, will default to the majority-class prediction thus producing misleadingly high accuracy. Balanced weights force models to attend to all classes and produce realistic and linguistically interpretable results.

9 Limitations

Several limitations were experienced and warrant mention. First, the corpus is admittedly small, with only 390 total instances, with only 121 training and 30 test instances available for layers 2 and 3. Rare categories such as ATTACK_QUESTIONER cannot yet be reliably evaluated using conventional models at such small scale. It is also worth noting that a faulty data spreadsheet resulted in several instances not being evaluated.

Second, as mentioned in the discussion section, the annotator pool was problematic in its design.

Third, the nature of the partial overlap design means that no item was rated by all annotators, this makes agreement estimates imprecise (79, 32, and 31 overlapping items for layers 1-3 respectively).

Fourth, the multi-label design for layers 2 and 3 increases annotation complexity and therefore the likelihood of reduced agreement among annotators relative to the single-label design constraints of layer 1.

Fifth, the corpus produced is limited in scope with respect to the availability of U.S. presidential debates available.

Lastley, The corpus is limited to U.S. presidential elections. It is plausible that incorporating a more varied collection of debates may yield different results.

10 Future Considerations

The most pressing issue is a guideline revision for layers 2 and 3. Clearer operational definitions, decision trees, worked examples may serve to yield more consistent results. An annotator training session is advisable as well as a calibration session where disagreements may be discussed and resolved before independent annotation commences. Homogenization of the annotator pool is also to be considered in order to more precisely control for apriori variables.

As for modeling, transformer-based encoders such as RoBERTa (Liu et al., 2019) or DeBERTa (He et al., 2021) in conjunction with added features such as type of question, may represent a more effective approach. A cross domain evaluation of Q&A transcripts may also serve to improve generalization of both annotation guidelines and of model predictions.

11 Conclusion

This paper has presented a pilot study of automated answerhood detection in presidential debate transcripts. It introduced a corpus of presidential debate Q&A pairs, a hierarchical three-layer annotation scheme, reported inter-annotator agreement across six annotators of varying backgrounds using a partial overlap design and established a TF-IDF logistic regression baseline with balanced class weights. The central finding is that classification performance track closely with annotator agreement: layer 1 ($\alpha 0.4471$, macro F1 = 0.70) yields reliable classification, while layers 2 and 3 require guideline refinements and a more rigorous annotator training protocol.

11.1 Appendix

11.1.1 Data

https://github.com/jgarciar83/Answerhood_in_Presidential_Debates_Corpus

11.1.2 The American Presidency Project

https://www.presidency.ucsb.edu/documents/presidential-documents-archive-guidebook/presidential-campaigns-debates-and-endorsements-0

```
Combined shape: (397, 9)
Dropping 6 row(s) with ambiguous 'ATTACK' label in layer_3.

==================================================
  Inter-Annotator Agreement
==================================================
  Unique annotator IDs found: 6 → ['AM', 'TM', 'MR', 'CLM', 'BD', 'A.A']

  Layer column : layer_1
  Annotators   : 6 → ['A.A', 'AM', 'BD', 'CLM', 'MR', 'TM']
  Items with ≥1 annotation : 213
  Items with all raters    : 0
  Items with ≥2 raters     : 79
  Krippendorff's Alpha     : 0.4471  (handles missing data)
  No fully overlapping items — computing pairwise Cohen's Kappa:
    A.A vs CLM: κ = 0.4299  (n=42)
    A.A vs MR: κ = 0.6199  (n=42)
    A.A vs TM: κ = 0.5695  (n=42)
    AM vs BD: κ = 0.2475  (n=37)
    CLM vs MR: κ = 0.4273  (n=42)
    CLM vs TM: κ = 0.5734  (n=42)
    MR vs TM: κ = 0.3837  (n=42)

  Layer column : layer_2
  Annotators   : 6 → ['A.A', 'AM', 'BD', 'CLM', 'MR', 'TM']
  Items with ≥1 annotation : 90
  Items with all raters    : 0
  Items with ≥2 raters     : 32
  Krippendorff's Alpha     : 0.1479  (handles missing data)
  No fully overlapping items — computing pairwise Cohen's Kappa:
    A.A vs CLM: κ = 0.3478  (n=15)
    A.A vs MR: κ = 0.4359  (n=11)
    A.A vs TM: κ = 0.1818  (n=18)
    AM vs BD: κ = 0.1667  (n=5)
    CLM vs MR: κ = 0.0000  (n=10)
    CLM vs TM: κ = -0.0687  (n=20)
    MR vs TM: κ = 0.3229  (n=13)

  Layer column : layer_3
  Annotators   : 6 → ['A.A', 'AM', 'BD', 'CLM', 'MR', 'TM']
  Items with ≥1 annotation : 80
  Items with all raters    : 0
  Items with ≥2 raters     : 31
  Krippendorff's Alpha     : 0.3558  (handles missing data)
  No fully overlapping items — computing pairwise Cohen's Kappa:
    A.A vs CLM: κ = 0.4545  (n=15)
    A.A vs MR: κ = 0.2453  (n=10)
    A.A vs TM: κ = 0.3684  (n=6)
    AM vs BD: κ = 0.0000  (n=4)
    CLM vs MR: κ = 0.0000  (n=12)
    CLM vs TM: κ = 0.0000  (n=8)
    MR vs TM: κ = 1.0000  (n=14)


Dropping 1 row(s) with no valid Layer 1 label.

NOT_DIRECT rows — train: 121, test: 30

---- Sanity check: y2 (NOT_DIRECT only) ----
  Empty lists   — train: 0, test: 0
  Non-list rows — train: 0, test: 0
  numpy ints    — train: 0, test: 0

---- Sanity check: y3 (NOT_DIRECT only) ----
  Empty lists   — train: 0, test: 0
  Non-list rows — train: 0, test: 0
  numpy ints    — train: 0, test: 0

==================================================
  Layer 1 Evaluation (DIRECT vs NOT_DIRECT)
==================================================
              precision    recall  f1-score   support

      DIRECT       0.73      0.89      0.80        46
  NOT_DIRECT       0.75      0.50      0.60        30

    accuracy                           0.74        76
   macro avg       0.74      0.70      0.70        76
weighted avg       0.74      0.74      0.72        76


==================================================
  Layer 2 Evaluation
==================================================
Classes present in test set: ['1', '2', '3']
              precision    recall  f1-score   support

           1       0.50      0.62      0.55        13
           2       0.38      0.38      0.38         8
           3       0.67      0.44      0.53         9

   micro avg       0.50      0.50      0.50        30
   macro avg       0.51      0.48      0.49        30
weighted avg       0.52      0.50      0.50        30
 samples avg       0.43      0.48      0.44        30
```

Figure 1: code output

```
==================================================
  Layer 3 Evaluation
==================================================
Classes present in test set: ['1', '2', '3']
              precision    recall  f1-score   support

           1       0.00      0.00      0.00         1
           2       0.55      0.67      0.60         9
           3       0.50      0.33      0.40        12

   micro avg       0.31      0.45      0.37        22
   macro avg       0.35      0.33      0.33        22
weighted avg       0.50      0.45      0.46        22
 samples avg       0.25      0.33      0.28        22
```

Figure 2: code output continued

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